Windows Server Network Troubleshooting Lab

Project Summary

Built a Windows Server troubleshooting lab in VirtualBox to investigate a no-internet issue. Reviewed browser behavior, adapter status, network details, Windows diagnostics, and TCP/IPv4 configuration to identify the root cause.

Skills Demonstrated

Windows Server troubleshooting, network connectivity analysis, DHCP review, IPv4 configuration review, root cause analysis, technical documentation

Objective:

Diagnose and investigate a no-internet issue in a Windows Server virtual machine by reviewing browser behavior, checking network configuration, validating adapter status, and documenting the troubleshooting process.

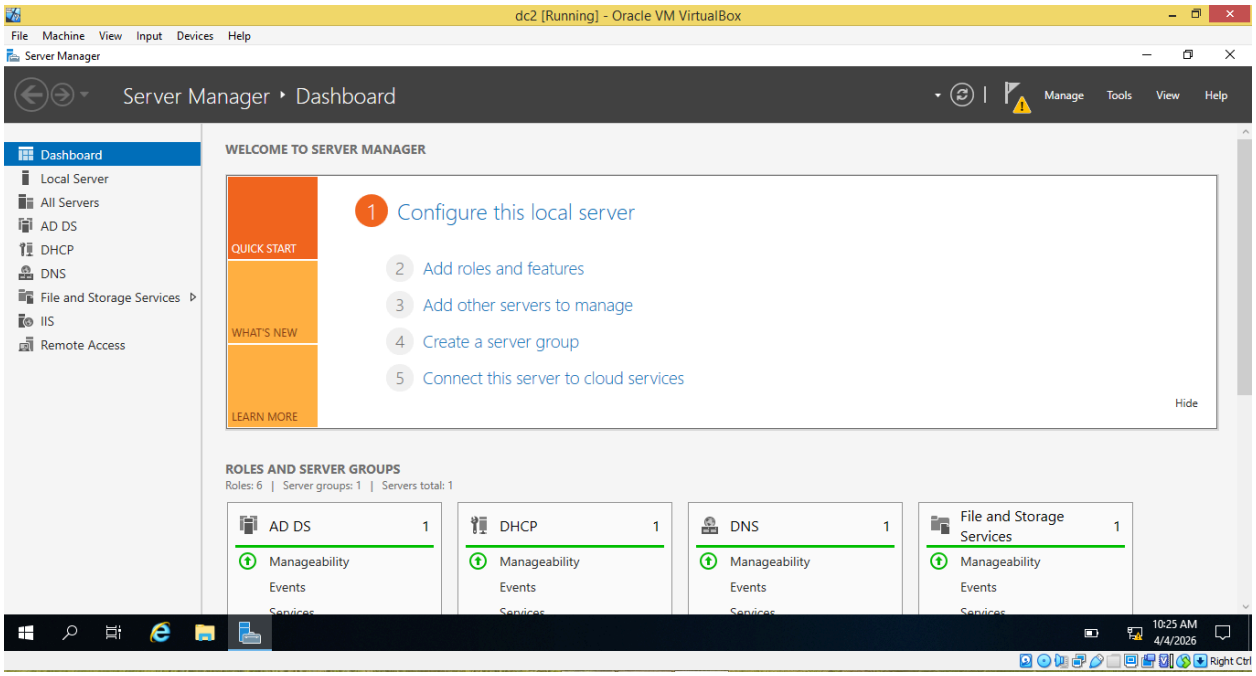
Tools Used:

VirtualBox, Windows Server, Server Manager, Internet Explorer, Network and Sharing Center, Network Connections, Adapter Status, TCP/IPv4 Properties

Environment Setup:

One Windows Server virtual machine configured in VirtualBox to simulate a Tier 1 help desk and server-support network troubleshooting scenario.

Server Manager Dashboard Screenshot



Problem

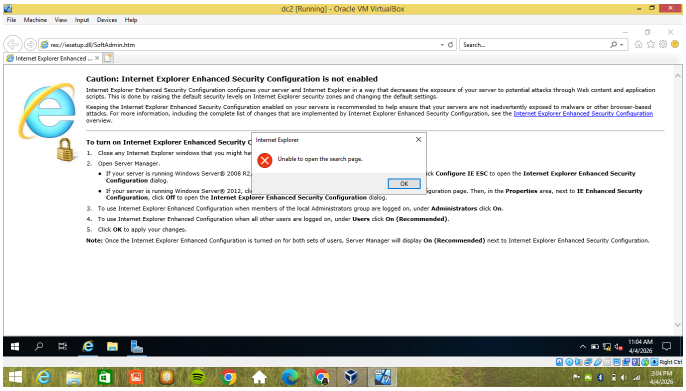
A Windows Server virtual machine could not access external web content. This issue prevented normal browser-based access and pointed to a broader network connectivity problem. The goal of this lab was to isolate the cause of the failure and document the troubleshooting process used to identify the root issue.

Steps Taken

1. Opened the Windows Server virtual machine and verified the environment from the Server Manager dashboard.
2. Opened Internet Explorer and attempted to access external web content.
3. Observed an “Unable to open the search page” error.
4. Opened Network and Sharing Center to review overall connection status.
5. Confirmed that the active network showed No Internet access.
6. Reviewed Network Connections and identified MY INTERNET as the relevant adapter for external connectivity.
7. Opened the adapter status window and confirmed that IPv4 and IPv6 both showed no network access while the adapter remained enabled.
8. Opened the connection details window and reviewed the IP configuration.
9. Ran Windows Network Diagnostics and confirmed that DHCP was not enabled for the adapter.
10. Opened the TCP/IPv4 properties and confirmed that the adapter was manually configured with a static IP address, no default gateway, and a loopback DNS server.

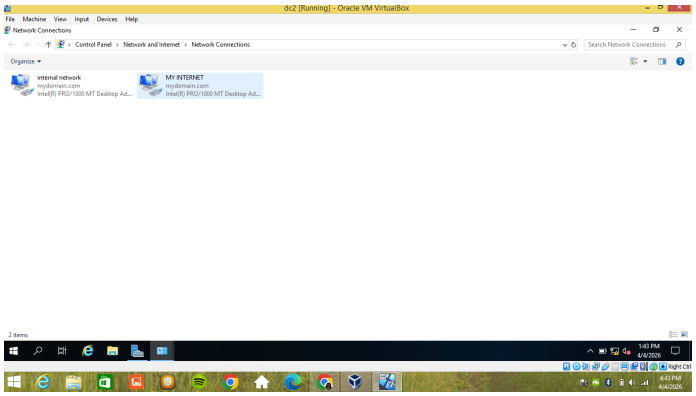
Evidence 1: Initial Browser Connectivity Failure

Internet Explorer returned an “Unable to open the search page” error, confirming that the virtual machine could not successfully access external web content.



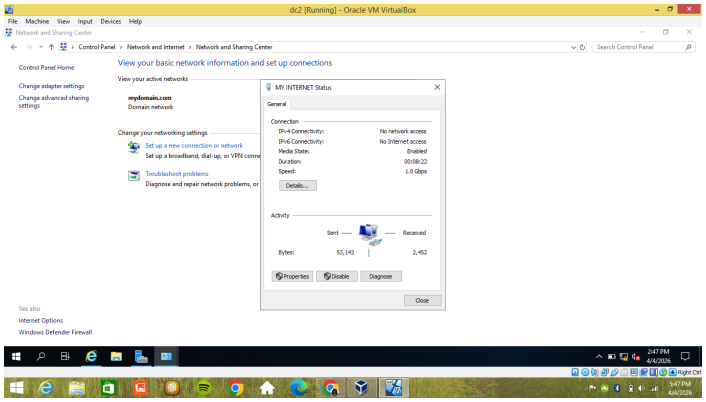
Evidence 2: Network Adapter Status

The Network Connections window was reviewed during troubleshooting. The virtual machine showed multiple network adapters, including an internal network adapter and an adapter labeled **MY INTERNET**, confirming that network interfaces were present and available for further investigation.



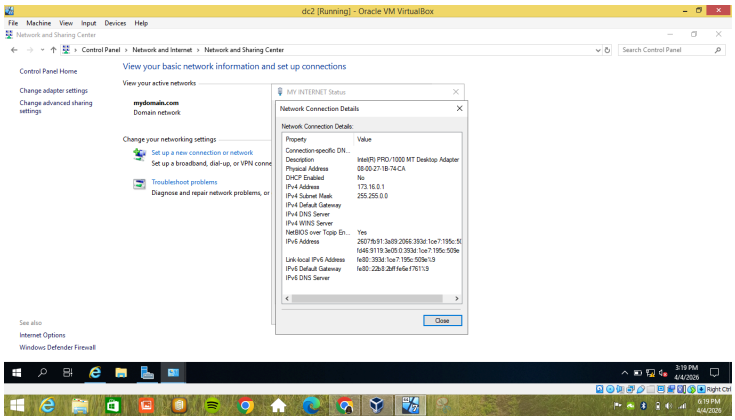
Evidence 3: Adapter Status Showing No Network Access

The **MY INTERNET** adapter status was reviewed during troubleshooting. The adapter was enabled, but both **IPv4 Connectivity** and **IPv6 Connectivity** showed no network access, confirming that the issue was not caused by the adapter being turned off.



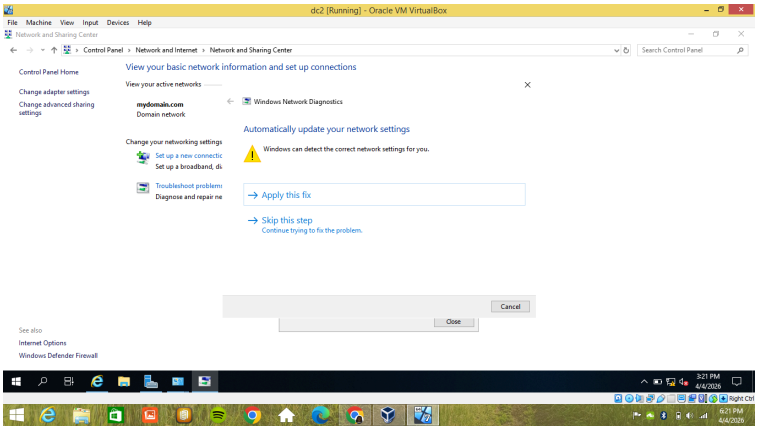
Evidence 4: Network Connection Details

The connection details for the **MY INTERNET** adapter were reviewed to inspect IP configuration, gateway information, and related network settings. The adapter showed a static IPv4 address of **173.16.0.1** with no visible IPv4 default gateway, which suggested a likely configuration issue affecting external connectivity.



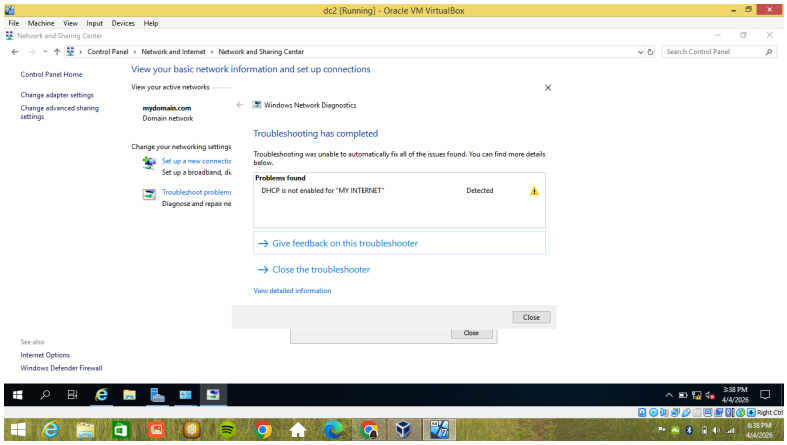
Evidence 5: Windows Network Diagnostic Recommendation

The built-in Windows Network Diagnostics tool was launched during troubleshooting. Windows identified a possible network settings issue and recommended applying an automatic fix.



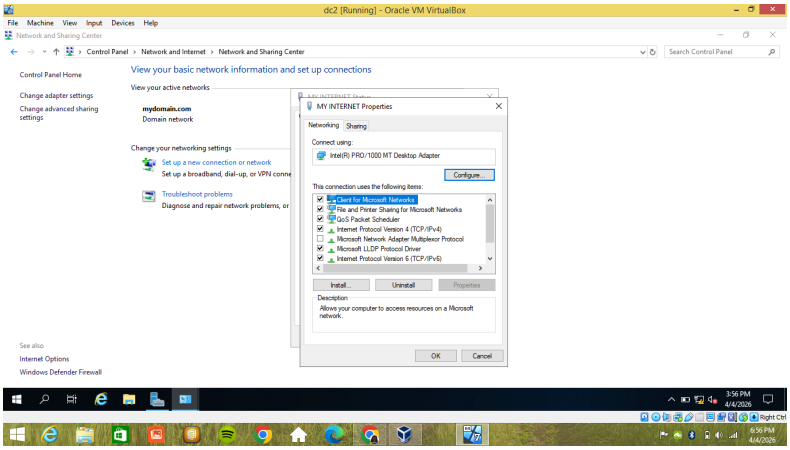
Evidence 6: Diagnostic Result Identifying DHCP Configuration Issue

The Windows Network Diagnostics tool completed its analysis and reported that DHCP was not enabled for the “MY INTERNET” adapter. This result indicated that the adapter was using a static configuration rather than automatically receiving network settings, which likely contributed to the lack of external connectivity.



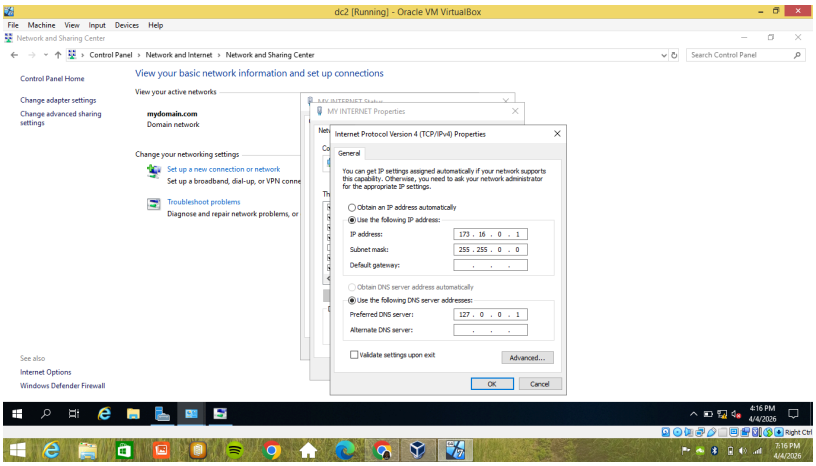
Evidence 7: Adapter Properties Review

The properties of the “MY INTERNET” adapter were reviewed to inspect the network components and confirm that Internet Protocol Version 4 (TCP/IPv4) was enabled as part of the troubleshooting process.



Evidence 8: IPv4 Configuration Review

The TCP/IPv4 configuration for the “MY INTERNET” adapter was reviewed during troubleshooting. The adapter was manually configured with a static IP address (`173.16.0.1`), subnet mask (`255.255.0.0`), no default gateway, and a preferred DNS server of `127.0.0.1`, which explains why the virtual machine could not reach external web content.



Current Finding

The troubleshooting process confirmed that the virtual machine’s connectivity issue was caused by incorrect manual IPv4 settings on the **MY INTERNET** adapter. Although the adapter was enabled, it was configured with a static IP address, no default gateway, and a loopback DNS server, which prevented external network access.

Resolution / Outcome

The investigation identified that DHCP was not enabled and that the adapter was using

manual IPv4 settings that were not valid for external connectivity. This documented the root cause of the no-internet issue in the virtual machine.

What I Learned

I learned how to investigate a Windows Server network connectivity issue by validating browser failure, reviewing network overview status, checking adapter state, inspecting connection details, running Windows diagnostics, and identifying incorrect IPv4 configuration as the root cause.